Junxuan Liang

whitefork@sjtu.edu.cn https://hitefork.github.io hitefork

RESEARCH INTERESTS

My research interests mainly lie in **Embodied AI**. I am now focusing on **human motion generation and human motion control**. I also have a background in human motion analysis, human object interaction.

EDUCATION

SEIEE, Shanghai Jiao Tong University — Shanghai, China
B.Eng. in Computer Science, Zhiyuan Honors Program of Engineering — Sept 2022 – Present

PUBLICATIONS

Efficient and Scalable Monocular Human-Object Interaction Motion Reconstruction
Boran Wen, Ye Lu, Keyan Wan, Sirui Wang, Jiahong Zhou, Junxuan Liang, Xinpeng Liu, Bang Xiao, Dingbang Huang, Ruiyang Liu, Yong-Lu Li*.
Under review. [wenboran2002.github.io/open4dhoi/] [Arxiv:2512.00960]
Explore physics-based whole-body human object interaction imitation across diverse objects and skills

ImDy: Human Inverse Dynamics from Imitated Observations
Xinpeng Liu, Junxuan Liang, Zili Lin, Haowen Hou, Yong-Lu Li*, Cewu Lu*.
ICLR, 2025. [foruck.github.io/ImDy/] [Arxiv:2410.17610]
Explore downstream tasks of human inverse dynamics in Addbiomechanics

Homogeneous Dynamics Space for Heterogeneous Humans
Xinpeng Liu, Junxuan Liang, Chenshuo Zhang, Zixuan Cai, Cewu Lu*, Yong-Lu Li*.
CVPR, 2025. [foruck.github.io/HDyS/] [Arxiv:2412.06146]
Explore downstream tasks of HDyS in Addbiomechanics

M^3-VOS: Multi-Phase, Multi-Transition, and Multi-Scenery Video Object Segmentation
Zixuan Chen, Jiaxin Li, Liming Tan, Yejie Guo, Junxuan Liang, Cewu Lu, Yong-Lu Li*.
CVPR, 2025. [M-cube-VOS.github.io/] [Arxiv:2412.13803]
Build M^3-VOS dataset

EXPERIENCE

Shanghai Jiao Tong University - XHumanLab — Shanghai, China
Undergraduate Research Assistant - Supervisor: *Prof.* Yichao Yan — Sept 2025 – Present

- Explore physics-based whole-body human object interaction imitation across diverse objects and skills

Shanghai Jiao Tong University - Machine Vision and Intelligence Group — Shanghai, China
Undergraduate Research Assistant - Supervisor: *Prof.* Cewu Lu and *Prof.* Yong-Lu Li — Feb 2024 – Present

- Explore the recently progressive human motion imitation algorithms to train a data-driven human inverse dynamics solver in a fully supervised manner
- Explore a fundamental reusable space HDyS for human dynamics by unifying the heterogeneity
- Build a video object segmentation benchmark focused on object phase transitions

SELECTED AWARDS

Zhiyuan Honors Program Scholarship, SJTU, Shanghai, China	2023
Third-Class Scholarship, SJTU, Shanghai, China	2024
Academic Progress Scholarship, SJTU, Shanghai, China	2025
Merit Student (top 3%), SJTU, Shanghai, China	2024
Meritorious Winner of Mathematical Contest In Modelling (Top 6.6%), America	2024
Outstanding Winner of Electronic&Electrical Engineering Innovation Competition, China	2024

SKILLS

- **Languages:** English (CET6 528, IELTS 6.5), Native Cantonese & Mandarin
- **Programming:** Python(Pytorch), C/C++, Linux, Nimble Physics, Latex, Typst